LABORATORY RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Week No.:	4
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Date:	30-08-2026

1. TITLE

Develop Dynamic Student Management System Webpage Using JavaScript DOM

2. AIM

To build an interactive dynamic student management webpage with real-time DOM table insertion and input validation using JavaScript.

3. OBJECTIVE

- Implement DOM event listeners for button clicks.
- Validate form inputs (name, age, course) before table insertion.
- Dynamically create table row and cell elements (tr, td) and append to table body.

4. THEORY

JavaScript event listeners and DOM manipulation enable real-time UI interactions. By responding to user click events and constructing table nodes dynamically, web applications can manage tabular records without reloading.

5. ALGORITHM / PROCEDURE

1. Create index.html, style.css, and script.js files.
2. Build form with Name, Age, and Course input fields and Add Student button.
3. Create table with Name, Age, and Course headers.
4. Attach click event listener to Add Student button.
5. Validate inputs; show error message if any field is empty.
6. Create tr and td elements, populate text, and append to student table.
7. Test in web browser.

6. PROGRAM

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Student Management</title>

    <link rel="stylesheet" href="style.css">
</head>

<body>

    <div class="container">

        <h1>Student Management System</h1>

        <div class="form">
            <input type="text" id="studentName" placeholder="Enter Name">
            <input type="number" id="studentAge" placeholder="Enter Age">

            <input type="text" id="studentCourse" placeholder="Enter Course">

            <button id="addStudent">Add Student</button>
        </div>

        <p id="message"></p>

        <table>

            <thead>
                <tr>
                    <th>Name</th>
                    <th>Age</th>
                    <th>Course</th>
                </tr>
            </thead>

            <tbody id="studentTable">
                <!-- JavaScript will add students here -->
            </tbody>

        </table>

    </div>

    <script src="script.js"></script>
</body>

</html>
```

style.css

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;

    background: linear-gradient(135deg, #667eea, #764ba2);

    min-height: 100vh;

    display: flex;
    justify-content: center;
    align-items: center;

    padding: 30px;
}

.container {
    width: 700px;

    background-color: white;
    padding: 35px;

    border-radius: 15px;

    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.25);
}

h1 {
    text-align: center;

    color: #333;

    margin-bottom: 30px;
}

.form {
    display: flex;
    flex-direction: column;
    align-items: center;

    gap: 15px;
}

.form input {
    width: 300px;
    padding: 12px;

    border: 2px solid #ddd;

    border-radius: 8px;

    font-size: 15px;

    outline: none;
}

.form input:focus {
    border-color: #667eea;
}
```

```
.form button {
    width: 150px;

    padding: 12px;

    border: none;

    border-radius: 8px;
    background-color: #667eea;

    color: white;

    font-size: 15px;

    cursor: pointer;
}

.form button:hover {
    background-color: #4f63c7;
}

#message {
    margin-top: 15px;

    text-align: center;

    color: #e74c3c;

    font-weight: bold;
}

table {
    width: 100%;
    margin-top: 30px;

    border-collapse: collapse;
}

th {
    background-color: #667eea;

    color: white;

    padding: 12px;
}

td {
    border: 1px solid #ddd;

    padding: 12px;

    text-align: center;
}

tr:nth-child(even) {
    background-color: #f5f6ff;
}
tr:hover {
    background-color: #eef0ff;
}

td button {
    padding: 7px 12px;
    border: none;
```

```
    border-radius: 6px;

    background-color: #e74c3c;

    color: white;

    cursor: pointer;
}

td button:hover {
    background-color: #c0392b;
}
```

script.js

```
let nameInput = document.getElementById("studentName");
let ageInput = document.getElementById("studentAge");
let courseInput = document.getElementById("studentCourse");

let addButton = document.getElementById("addStudent");

let studentTable = document.getElementById("studentTable");
let message = document.getElementById("message");


addButton.addEventListener("click", function() {

    let name = nameInput.value;
    let age = ageInput.value;
    let course = courseInput.value;


    if (name === "" || age === "" || course === "") {

        message.textContent = "Please fill all fields";

        return;
    }
    let row = document.createElement("tr");

    let nameCell = document.createElement("td");
    let ageCell = document.createElement("td");
    let courseCell = document.createElement("td");


    nameCell.textContent = name;
    ageCell.textContent = age;
    courseCell.textContent = course;


    row.appendChild(nameCell);
    row.appendChild(ageCell);
    row.appendChild(courseCell);


    studentTable.appendChild(row);

});
```

7. CODE EXPLANATION

Code / Syntax	Explanation
addButton.addEventListener('click', ...)	Attaches event handler to process student addition
document.createElement('tr')	Creates dynamic table row element
row.appendChild(nameCell)	Attaches data cells to newly constructed table row
studentTable.appendChild(row)	Inserts completed row into live HTML table body

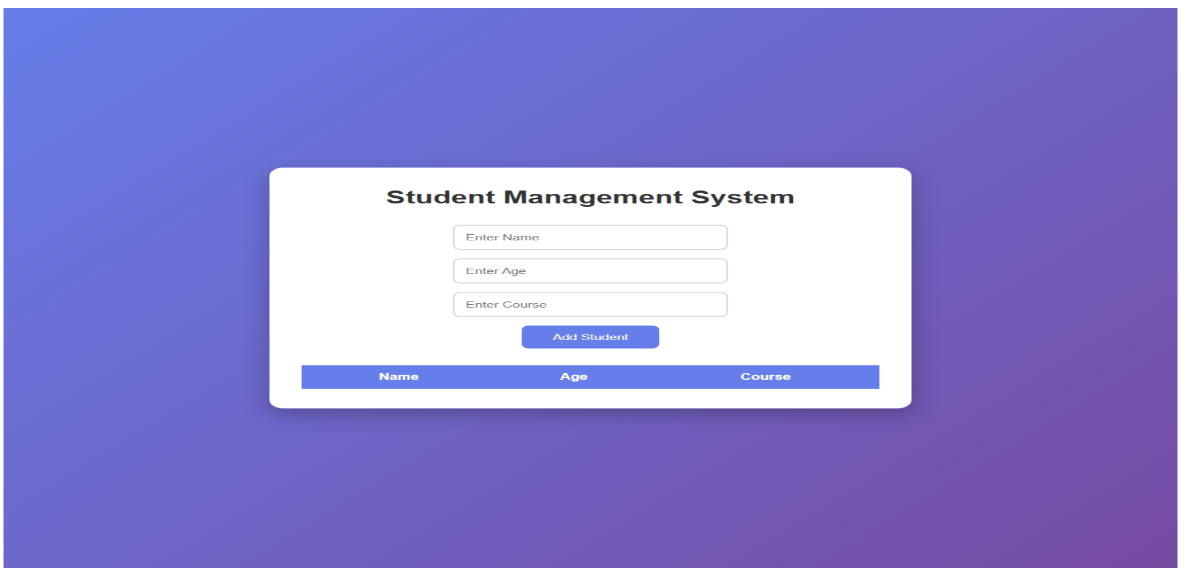
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Form Validation	Click Add with empty fields	Error message 'Please fill all fields' displayed	**Pass**
TC-02	Add Record	Enter student details and click Add	New row added to student table dynamically	**Pass**

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Week 4 (Develop Dynamic Student Management System Webpage Using JavaScript DOM) was successfully executed and verified.